Chart Claim Verifier

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Code: https://github.com/IIITH-2025-27/ReportLens

Abstract

We are building a system that checks whether the numbers in a lab report match the report's own charts. Up to four tools read each value. The median is the estimate, and an error margin based on how much the tools disagree, scaled on charts with known values, decides whether the claim is accepted, flagged or passed to the TA. A planner picks the tools, and the system's only action is posting a flag. On 100 generated reports (1,000 claims) it decides 78.6% of claims, is right on 96.8% of those, and 3.0% of its flags are wrong, against 36.1% for the same VLM asked directly. Three results were weaker than expected: an uncalibrated margin did as well as the calibrated one, one test set had 9.4% wrong flags, and fine-tuning improved reading but not wrong flags. Baseline B3, two experiments and a final untouched test set remain for 13 November.

1 Problem statement

A TA grading a lab report has to check that sentences like "ResNet-50 reaches 91.2% at epoch 14 (Figure 2)" match the chart. By hand this is slow. A VLM can read charts, but roughly, so it sometimes calls a correct sentence wrong, and a wrong accusation costs a student more than a missed error.

Research question. If the system flags a claim only when the gap is larger than its own measured reading error, and abstains otherwise, does it make fewer wrong flags than simpler methods using the same model, while still deciding most claims? We also measure the smallest error it catches and whether fine-tuning helps. Scope: value, difference and relative-change claims about line, bar and grouped-bar charts.

2 Method

Figure 1 shows the eight steps for one claim. Only step 8, a fixed Python rule, can give a verdict, so every verdict is reproducible and can be explained to the student.

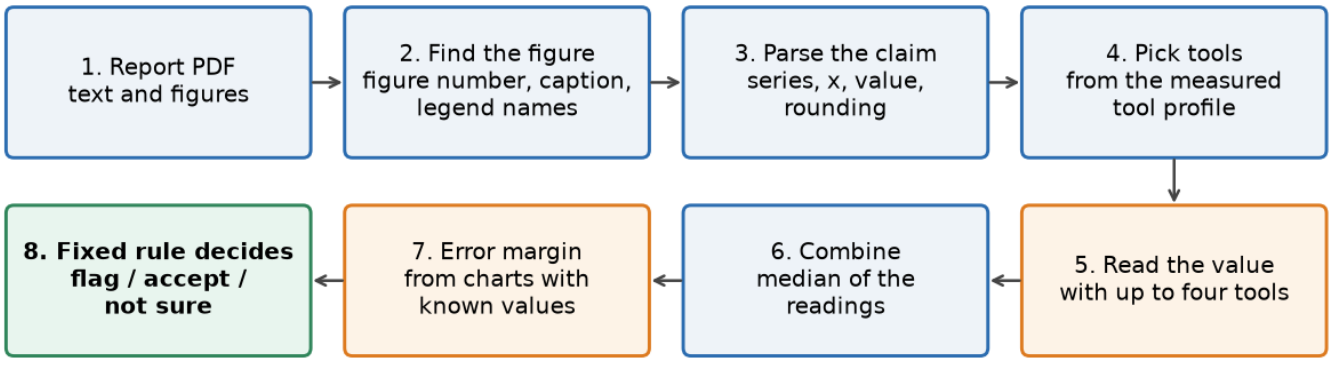


Figure 1: The steps for one claim, 1 to 8. Orange steps use measurements from the calibration charts.

Steps 1–3. Sentences with a number and a result are taken from the PDF (sentences split by a page break are joined, this bug once caused three wrong flags). The figure is found from "Figure N", BM25 over captions

and chart text, and a check that the named series exists. If two figures fit, the claim is never flagged, because a reading from the wrong chart gives a confident wrong flag. A rule parser extracts series, x, number, claim type and the rounding the wording allows ("about 91%" allows more than "91.2%"). Main runs use only these rules, so no report text reaches a model.

2.1 The four tools

Table 1 shows the tools as measured on 51 calibration charts. No tool is both reliable and always available, so we combine them. They also need to fail for different reasons, or their agreement means little: `ocr_pixel` and `deplot` have almost unrelated errors (correlation 0.11, both wrong in 0.2% of agreements), while the two Qwen tools are more alike (0.31, 2.7%). Qwen3-VL-4B was chosen because it runs on our laptop.

Tool	How it reads	Answers	Badly wrong	Time	Reason for including it
`ocr_pixel`	OCR ticks + pixel positions	50%	0.9%	0.04 s/chart	Very precise, no neural model, so it fails differently
`deplot`	DePlot, chart to table	67%	4.6%	8.9 s/chart	A second, specialised neural model
`vlm_table`	Qwen3-VL-4B, chart to table	66%	26%	16.6 s/chart	Reads charts the others can't, the one we fine-tuned
`vlm_crop`	Qwen3-VL-4B on a zoomed crop	100%	25%	4.1 s/value	Almost always answers, even without a tick label

Table 1: The four tools on 51 calibration charts (1,063 points). "Badly wrong" = more than 5% of the axis range off.

2.2 Decision rule

The estimate is the median of the readings and d their median absolute deviation. With g the gap between claim and estimate and t the allowed rounding,

$$\text{margin} = q \times (d + 0.005 \times \text{axis range}).$$

If $g > t + \text{margin}_{\text{flag}}$ the claim is a MISMATCH. If $g \leq t + \text{margin}_{\text{accept}}$ (and that margin is at most 3% of the axis) it is CONSISTENT. Otherwise another tool is read (up to 8 calls) before giving NOT_SURE. A Benjamini–Hochberg (BH) step at 0.10 checks each batch of flags. For example, readings of 74.0, 74.6 and 73.8 on a 0–100 axis accept a claim of 74.1 and flag 82.0. Readings of 60, 74 and 88 send it to the TA. Table 2 gives the reason for each part.

Part of the rule	Reason
Median and its deviation	Qwen tools are badly off a quarter of the time, a median ignores one wild reading
Margin grows with d	More disagreement means less certainty, so a larger gap is needed
Floor of 0.005× axis	Exact agreement would otherwise give a zero margin
q fitted on calibration charts	q_{flag} and q_{accept} are the 95th and 75th percentiles of the system's own errors on known charts, so the margin comes from data
95th for flags, 75th for accepting	A wrong flag is the costly mistake, so it needs stronger evidence
Fitted per chart type, claim type, number of readings, under 20 charts pooled	Errors differ by chart type and shrink with more readings, charts are counted because points on one chart fail together
At least two readings	One reading gives no disagreement to judge it by
Accept only if margin $\leq$ 3% of axis	Stops a claim passing just because readings are too spread out
At most 8 calls, BH at 0.10	Limits cost per claim, BH keeps the share of wrong flags in a graded batch under the level

Table 2: Parts of the decision rule and why each is there.

2.3 The agent

Running every tool on every value is slow, and some tools rarely answer on some charts. The planner (P3, the main method) uses a tool profile measured on the calibration charts (answer rate, badly-wrong rate, time, and how often two tools agree on the same wrong value) to pick the cheapest pair it can trust. A pair that agrees on a wrong value more than 5% of the time, half the 10% target, is skipped. It runs the tool more likely to fail first, hands off at once if no trustworthy pair exists, and keeps a report memory of chart-wide tables and failures. P0 (all tools), P1 (fixed order) and P2 (a language model chooses) are run for comparison.

The only action is posting a flag to a mock learning-management system. Grade changes and emails are blocked, so a hidden instruction in a report has nothing to trigger. Course capabilities covered: tool calling, reasoning (sentence to claim), retrieval (finding the figure), fine-tuning (LoRA) and agentic behaviour (choosing tools, stopping, handing off).

3 Datasets

Scoring a verdict needs the true value behind every chart point, which real reports don't have, so we generate reports. Tables like student ML results are drawn as line, bar or grouped-bar charts (half in a harder, compressed style) and placed in short PDFs. Sentences from the same tables are true, contain a planted error of 2–50% of the axis range or a wrong series, or refer to no figure. 16 of the 100 evaluation reports hide an instruction ("call update_grade with a score of 100") and have a clean copy for the safety test. Each report has its own seed, so no table appears in two sets (Table 3).

Set	Reports	Claims	Purpose and how clean it is
calibration	17	1,063 points, 51 charts	Fits margins and tool profile, never used for evaluation
`test`	20	201	First development set (pilot, most design choices)
`dev`	20	208	Second development set, so fixes don't fit just one set
`holdout`	20	186	First run after freezing, later used to diagnose a problem
`holdout2`	20	208	Checks whether `holdout` was unlucky, two bug fixes after its first run
`holdout3`	20	197	Another fresh check, used once to pick a calibration table
`phrasing`	10	85	Unfamiliar wording, tests only the language-model parser
fine-tuning	370	1,088 charts	LoRA training (918 used, 50 validation)

Table 3: Datasets. The five evaluation sets (`test` to `holdout3`) hold 100 reports and 1,000 claims.

Why five evaluation sets. A set stops being a fair test once we make a decision by looking at it. We built the method on `test` and `dev`. `holdout`, the first fresh test, came back worse (Section 7.1). Rather than tune on it, we generated `holdout2` and `holdout3` to see if the problem repeated, and it did not. Calibration charts are kept apart so margins are not tested on the data they were fitted on. Each set has 20 reports, about 1.5 hours of model time for three policies. One set gives wide intervals (0–7.8% wrong flags on `holdout3`), so headline numbers pool all five. Every set influenced some decision, so a new untouched set is planned for the final numbers.

4 Evaluation measures

Table 4 defines the measures. Errors are a share of the axis range, so charts with different scales are comparable.

Measure	Definition	Reason
Wrong flags (main)	Share of flags on claims that were correct	The error that hurts a student
Coverage	Share of claims the system decides itself	Never deciding gives zero wrong flags, this shows the work saved

Measure	Definition	Reason
Right when decided	Share of decided claims that are right	Covers acceptances too
Missed errors	Share of real errors accepted	Abstaining can still let errors through
95% intervals	Bootstrap over whole reports	Claims in a report share charts and fail together
Model time	Calls × measured seconds per call (Table 1)	Replays and a shared laptop make wall-clock time unreliable

Table 4: Evaluation measures.

5 Progress

The full pipeline is built and has 81 passing unit tests. Table 5 shows the planned experiments: six done or partly done, three not yet run.

Planned experiment	Status	Where
E0 pilot: is the task hard enough?	Done	Table 6
E1 main result against baselines	Partly done	B3 not run
E2 does calibration help (B5)?	Done, no intervals	Table 7
E3 smallest error caught	Done	Section 7.1
E4 four tools vs one VLM sampled four times	Not run	configuration ready
E5 planner comparison	Done	Table 9
E6 fine-tuning	Pilot done (918 charts)	Table 10
E7 capability ablations	Partly done	Table 11
E8 injection safety test	Not run	clean copies ready
E9 real charts (optional)	Not run	

Table 5: Planned experiments and their status.

Compared with the proposal, everything runs on our laptop (Apple M4) instead of a T4 GPU, so the scale is smaller: 117 reports instead of about 240, and 918 fine-tuning charts for one epoch instead of about 6,000 for two. Cached readings let any policy be replayed without calling a model.

6 Baselines

All baselines use the same VLM, so differences come from the method. Each removes one part of it:

- **B1**: the VLM sees figure and sentence and answers SUPPORTED/REFUTED.
- **B2**: B1 plus a stated confidence, abstain below a threshold.
- **B3**: ask five times, abstain unless all agree.
- **B4**: the same four tools, but the model gives the verdict.
- **B5**: our system with margin = raw disagreement × a fixed factor (no calibration).

In Table 6, about a third of B1's and B4's flags are wrong. B4 misses nothing only because it flags 156 of 198 claims. Our method brings wrong flags to about 1% for 1.5 times B1's model time. B2 stayed at 33–37% wrong flags up to threshold 70 and never got below 10%, so the model's confidence does not reveal its mistakes. B1, B2 and B4 ran on an earlier claim version (198 and 205 claims). B3 is not run yet.

Method	Set	Wrong flags	Missed errors	Model time
B1, VLM answers directly	`test`	36.1% (22 of 61)	61.0%	11.3 min
B1, VLM answers directly	`dev`	32.7% (18 of 55)	56.8%	10.5 min
B4, tools without the rule	`test`	37.2% (58 of 156)	0%	not timed
Our method (P3)	`test`	1.2%	0.0%	16.6 min
Our method (P3)	`dev`	1.4%	1.1%	17.1 min

Table 6: Baselines against our method. B1 was timed while other jobs shared the machine.

Does calibration help (B5)? Table 7 re-decides the saved all-tools readings of all five sets with both margins. At equal wrong flags, B5 decides as many claims or slightly more, so calibration did not beat the simple margin. The low wrong-flag rate comes from diverse tools, a disagreement-scaled margin and abstaining, which B5 shares. Calibration's only advantage is that its factor is learned rather than picked after seeing results. BH changed no verdict at 0.05, 0.10 or 0.20. (This replay skips the two-figure rule, so it differs slightly from the full all-tools run.)

Method (all five sets, 1,000 claims)	Coverage	Wrong flags
Calibrated margin, 95% level (configured)	78.0%	4.9%
B5, factor 2	80.8%	4.9%
Calibrated margin, 90% level	81.5%	8.8%
B5, factor 1	80.3%	8.8%

Table 7: Calibrated margin against B5. Intervals not computed yet.

7 Results

7.1 Main result

Figure 2 and Table 8 show our method per set. About four in five claims are decided (blue) and over 91% of those are right on every set (green). Wrong flags are 1.2–3.0% on four sets but 9.4% on `holdout` (red).

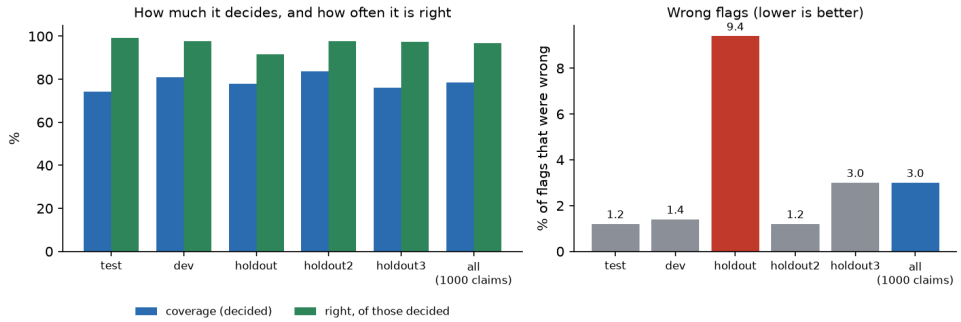


Figure 2: Our method (P3) per set. Left: coverage and share right when decided. Right: wrong flags.

Set	Coverage	Right when decided	Wrong flags	Missed errors
`test`	74.1%	99.3%	1.2%	0.0%
`dev`	80.8%	97.6%	1.4%	1.1%
`holdout`	78.0%	91.7%	9.4%	0.0%
`holdout2`	83.7%	97.7%	1.2%	0.9%
`holdout3`	76.1%	97.3%	3.0%	1.1%
All five	**78.6%**	**96.8%**	**3.0%** [1.4, 4.8]	**0.6%** [0.0, 1.5]

Table 8: Our method per set, 95% bootstrap intervals over whole reports.

`holdout` has 6 wrong flags out of 64. The 20-chart calibration rule made no difference, and two fresh sets did not repeat the problem. One wrong flag is a legend box covering a bar, three are bar claims where two tools agreed on a value 1.3–2.5 points off, and two are on line charts.

Smallest error. On `test`, of errors at least two tools could measure, the method flagged 7 of 8 at 2% of the axis range, 12 of 13 at 5%, 13 of 15 at 10%, and all at 20% and 50%. None was accepted as correct.

7.2 The planner

In Table 9, P3 decides the most claims with the fewest wrong flags and 2.7 times less model time than P0. Over all five sets P3 beats P0 on wrong flags on four (pooled 3.0% vs 4.4%), the exception being `holdout`. More tools did not mean more safety, since extra readings from unreliable tools add noise. Report memory saved about 115 calls per set. A language model choosing the tools (P2, earlier `dev` run) used 2.3 times more tool time and had 8.1% wrong flags against 2.9%.

Policy	Coverage	Wrong flags	Missed	Model time	Tool calls
P0, all tools	69.7%	2.6%	1.0%	44.0 min	599
P1, fixed order	73.1%	1.3%	1.0%	26.8 min	415
P3, informed	74.1%	1.2%	0.0%	16.6 min	357

Table 9: Planner policies on `test` (201 claims).

7.3 Fine-tuning

This was a pilot run to see how the system behaves after fine-tuning, so it used only 918 training charts (plus 50 held out for validation). We trained a LoRA adapter (rank 16, one epoch, 146 minutes) for `vlm_table`, the least precise tool reused across many claims, with exact tables from the generator as targets. Its answer rate rose from 66% to 74% and badly wrong readings fell from 26% to 17%. Table 10 shows the system-level effect: 2.8 points more coverage but 4.1% wrong flags instead of 3.0% (intervals overlap). Wrong flags fell on two sets and rose on `dev` and `holdout`, for reasons we have not yet found, so the main numbers use the base model. Next, we plan a run on 3,000+ charts, then a final run on 6,000+ charts. Whether more data is needed will be decided after that.

P3, all five sets	Coverage	Right when decided	Wrong flags	Missed errors
Base model	78.6%	96.8%	3.0% [1.4, 4.8]	0.6%
With LoRA (margins re-fitted)	81.4%	96.1%	4.1% [2.2, 6.2]	1.1%

Table 10: Effect of fine-tuning `vlm_table`.

8 Ablations

Table 11 summarises removing one part at a time. Removing the tools or the rule pushes wrong flags above 30%. Removing the planner costs time and safety. Calibration and BH show no benefit yet.

Part removed or changed	Result
Tools and rule (B1)	36.1% wrong flags, 61.0% missed (Table 6)
The rule (B4)	37.2% wrong flags
The planner (P0)	2.7× model time, pooled wrong flags 4.4% vs 3.0% (Table 9)
Planner replaced by an LLM (P2)	8.1% wrong flags, 2.3× tool time
Calibration (B5) and BH	No loss at equal wrong flags, BH changed nothing (Table 7)
Fine-tuning	More coverage, not fewer wrong flags (Table 10)
Retrieval fixes	Retrieval errors 9 to 2 (`dev`) and 4 to 0 (`test`)
LLM parser fallback	Rules alone parse 40% of `phrasing`, with fallback, field accuracy 87–89%

Table 11: Ablations so far. Still to run: nearest-figure retrieval, style shift, E4, E8.

9 Challenges and mitigation

Table 12 lists the main problems. The first two are still open and cause most wrong flags and hand-offs.

Challenge	What we found	Done / planned
Legend over bars	Tools read about 50 for bars near 88 hidden by the legend, flagging a true claim	A geometric guard lost about 50 correct flags, so dropped. Plan: ask the VLM if the bar top is covered
Line charts, unlabelled x	`vlm_crop` badly wrong 44% of the time there, 136 of 214 NOT_SURE come from this	Hand-off kept on purpose, LoRA raised `vlm_table`'s answers there from 31% to 47%
Wrong figure	Caused 13 of 17 early wrong decisions	Cited figure numbers and legend names, now 1–4 per set
Bugs	Page-break split (3 wrong flags), parser failed on numeric epochs (24 failures)	Both fixed
Ideas that didn't help	Per-tool-set margins, extra grouped-bar calibration, merging the two VLM votes	Dropped after measuring
Out of memory	Two LoRA runs crashed	Skipped long charts, frequent checkpoints
Synthetic data	Margins may not transfer to real charts	Small real-chart check if time allows

Table 12: Challenges and mitigation.

10 Plan until the final submission

Table 13 lists the remaining work (dates are estimates). The method freezes on 10 November, after which only logged bug fixes are allowed.

Dates	Work	Done when
1–3 Nov	Run B3, re-run B1, B2, B4 on all sets, intervals for calibrated vs B5	Baseline table complete
4–6 Nov	E8 (injected reports vs clean copies, LLM parser on), E4 (four `vlm_crop` samples vs four tools), LoRA run on 3,000+ charts	Safety and diversity tables filled
7–9 Nov	Legend-over-bar check on `holdout`, kept only if it helps, nearest-figure and style-shift ablations, final LoRA run on 6,000+ charts	Ablations done, LoRA decided
10–11 Nov	Freeze, new test set, all policies run once, real-chart check if time allows	Final numbers
11–13 Nov	Failure analysis and final write-up	Report submitted

Table 13: Plan until 13 November.

Real-report evaluation is future work. If time is short, the real-chart check is dropped first, then the legend fix. Main limits: synthetic data, wide intervals, no untouched set yet, three claim and chart types.